

Variational principle:

Variational principle gives an upper bound for the ground state energy, when Schrodinger eqn cannot be directly solved.

* Pick any normalized function ψ .

* The ground state energy E_{gs} is

$$E_{gs} \leq \langle \psi | H | \psi \rangle \equiv \langle H \rangle$$

* An appropriate choice of the trial function ψ will give E_{gs} that is close to actual value.

proof: $H\psi_n = E_n\psi_n$

$$\psi = \sum_n c_n \psi_n$$

$$\begin{aligned} \langle \psi | \psi \rangle = 1 &= \left\langle \sum_m c_m \psi_m \left| \sum_n c_n \psi_n \right. \right\rangle \\ &= \sum_m \sum_n c_m^* c_n \langle \psi_m | \psi_n \rangle \\ &= \sum_m \sum_n c_m^* c_n \delta_{mn} \\ &= \sum_n |c_n|^2 \end{aligned}$$

$$\begin{aligned} \langle H \rangle &= \left\langle \sum_m c_m \psi_m \left| H \sum_n c_n \psi_n \right. \right\rangle \\ &= \sum_m \sum_n \langle c_m \psi_m | c_n E_n \psi_n \rangle \quad [\because H\psi_n = E_n\psi_n] \\ &= \sum_m \sum_n c_m^* c_n E_n \delta_{mn} \\ &= \sum_n E_n |c_n|^2 \end{aligned}$$

But $E_{gs} \leq E_n$ since the ground state has the lowest eigenvalue.

$$\Rightarrow \langle \psi | H | \psi \rangle = \sum_n E_n |c_n|^2 \geq E_{gs} \underbrace{\sum_n |c_n|^2}_1$$

$$\Rightarrow \boxed{\langle \psi | H | \psi \rangle \geq E_{gs}}$$

1D Harmonic oscillator

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2$$

Use trial function $\psi(x) = A e^{-bx^2}$

Normalize $\psi(x)$

$$\begin{aligned}\langle \psi | \psi \rangle = 1 &= |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2} dx \\ &= 2|A|^2 \int_0^{\infty} e^{-2bx^2} dx\end{aligned}$$

$$\text{Now } \int_0^{\infty} x^{2n} e^{-x^2/a^2} dx = \sqrt{\pi} \frac{(2n)!}{n!} \left(\frac{a}{2}\right)^{2n+1}$$

$$\text{Here } n=0, \quad a = \sqrt{\frac{1}{2b}}$$

$$\langle \psi | \psi \rangle = 2|A|^2 \sqrt{\pi} \frac{1}{\sqrt{2b}} \frac{1}{2} = \left(\frac{\pi}{2b}\right)^{1/2} |A|^2 = 1$$

$$\Rightarrow A = \left(\frac{2b}{\pi}\right)^{1/4}$$

calculate $\langle \psi | H | \psi \rangle$ to get an upper bound for E_0

$$\langle H \rangle = \langle T \rangle + \langle V \rangle$$

$$\begin{aligned}\langle T \rangle &= -\frac{\hbar^2}{2m} |A|^2 \int_{-\infty}^{\infty} e^{-bx^2} \frac{d^2}{dx^2} (e^{-bx^2}) dx \\ &= -\frac{\hbar^2}{2m} |A|^2 \int_{-\infty}^{\infty} e^{-bx^2} (-b) \cdot 2 \frac{d}{dx} (e^{-bx^2} x) dx \\ &= 2b \cdot \frac{\hbar^2}{2m} \left(\frac{2b}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} e^{-bx^2} [e^{-bx^2} (-2bx^2) + e^{-bx^2}] dx \\ &= 2 \cdot 2b \frac{\hbar^2}{2m} \left(\frac{2b}{\pi}\right)^{1/2} \left[\int_0^{\infty} (-2bx^2 e^{-2bx^2}) dx + \int_0^{\infty} e^{-2bx^2} dx \right] \\ &= 4b \frac{\hbar^2}{2m} \left(\frac{2b}{\pi}\right)^{1/2} \left[-2b \frac{1}{4} \frac{1}{2b} \sqrt{\frac{\pi}{2b}} + \frac{1}{2} \sqrt{\frac{\pi}{2b}} \right] \\ &= 4b \frac{\hbar^2}{2m} \left(-\frac{1}{4} + \frac{1}{2}\right) = \frac{\hbar^2 b}{2m}\end{aligned}$$

$$\int_0^{\infty} x^2 e^{-2bx^2} dx = \sqrt{\pi} \frac{2}{1} \left(\frac{1}{\sqrt{2b}} \frac{1}{2} \right)^3 = \frac{1}{4} \cdot \frac{1}{2b} \sqrt{\frac{\pi}{2b}}$$

where $n=1$, $a = \frac{1}{\sqrt{2b}}$

$$\int_0^{\infty} x^{2n} e^{-x^2/a^2} dx = \sqrt{\pi} \frac{(2n)!}{n!} \left(\frac{a}{2} \right)^{2n+1}$$

$$\boxed{\langle T \rangle = \frac{\hbar^2 b}{2m}}$$

$$\langle V \rangle = \frac{1}{2} m \omega^2 |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2} x^2 dx$$

$$= \frac{1}{2} m \omega^2 \sqrt{\frac{2b}{\pi}} \cdot \frac{1}{2} \cdot \frac{1}{2b} \sqrt{\frac{\pi}{2b}}$$

$$= \frac{m \omega^2}{8b} \Rightarrow \boxed{\langle V \rangle = \frac{m \omega^2}{8b}}$$

$$\langle H \rangle = \langle T \rangle + \langle V \rangle = \frac{\hbar^2 b}{2m} + \frac{m \omega^2}{8b} \geq E_{gs} \text{ for any } b$$

minimize the expression w.r.t the parameter b

$$\frac{d}{db} \langle H \rangle = \frac{d}{db} \left(\frac{\hbar^2 b}{2m} + \frac{m \omega^2}{8b} \right) = 0$$

$$\Rightarrow \frac{\hbar^2}{2m} - \frac{m \omega^2}{8b^2} = 0 \Rightarrow b^2 = \frac{m^2 \omega^2}{4 \hbar^2}$$

$$\Rightarrow \boxed{b = \frac{m \omega}{2 \hbar}}$$

$$\Rightarrow \langle H \rangle = \frac{\hbar^2}{2m} \cdot \frac{m \omega}{2 \hbar} + \frac{m \omega^2}{8} \cdot \frac{2 \hbar}{m \omega}$$

$$= \frac{1}{4} \hbar \omega + \frac{1}{4} \hbar \omega = \frac{1}{2} \hbar \omega$$

$$\Rightarrow \boxed{E_{gs} \geq \frac{1}{2} \hbar \omega}$$

delta function potential

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - \alpha \delta(x)$$

Use trial function $\psi(x) = \left(\frac{2b}{\pi}\right)^{1/4} e^{-bx^2}$

$$\langle T \rangle = \frac{\hbar^2 b}{2m} \quad (\text{Recall the result for harmonic oscillator})$$

$$\langle V \rangle = -\alpha \left(\frac{2b}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} e^{-2bx^2} \delta(x) dx$$

$$= -\alpha \sqrt{\frac{2b}{\pi}}$$

$$\langle H \rangle = \frac{\hbar^2 b}{2m} - \alpha \sqrt{\frac{2b}{\pi}}$$

$$\frac{d\langle H \rangle}{db} = \frac{\hbar^2}{2m} - \frac{\alpha}{\sqrt{2b\pi}} = 0 \Rightarrow \frac{\hbar^4}{4m^2} = \frac{\alpha^2}{2b\pi}$$

$$\Rightarrow b = \frac{\alpha^2}{2\pi} \cdot \frac{4m^2}{\hbar^4} = \frac{2m^2 \alpha^2}{\pi \hbar^4}$$

$$\begin{aligned} \langle H \rangle &= \frac{\hbar^2}{2m} \cdot \frac{2m^2 \alpha^2}{\pi \hbar^4} - \alpha \cdot \frac{2}{\pi} \cdot \frac{m\alpha}{\hbar^2} \\ &= \frac{m\alpha^2}{\pi \hbar^2} - 2 \frac{m\alpha^2}{\pi \hbar^2} = -\frac{m\alpha^2}{\pi \hbar^2} \end{aligned}$$

$$E_{gs} \leq -\frac{m\alpha^2}{\pi \hbar^2}$$

Exact result $E_{gs} = -\frac{m\alpha^2}{2\hbar^2}$

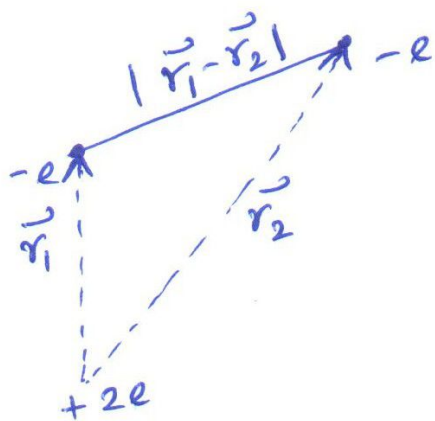
the ground state of Helium atom

Two electrons orbiting the nucleus, $z=2$

$$\hat{H} = -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{2e^2}{r_1} - \frac{2e^2}{r_2} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} \rightarrow \text{electron-electron repulsion term}$$

Experimentally obtained value of the ground state energy of helium $E_{gs} = -78.975 \text{ eV}$

The problem has no exact solution due to the introduction of e-e repulsion term in $\hat{H}$.



We ignore the e-e repulsion term

$\Rightarrow$ the problem reduces to two independent Hydrogen like Hamiltonians.

ground state $\rightarrow$ product of two hydrogen ground state wave functions with $z=2$

$$\Psi_0(\vec{r}_1, \vec{r}_2) = \Psi_{100}(\vec{r}_1) \Psi_{100}(\vec{r}_2)$$

$$\Psi_{100}(\vec{r}) = \sqrt{\frac{z^3}{\pi a_0^3}} e^{-zr/a_0} \quad z=2$$

$$\Rightarrow \Psi_0(\vec{r}_1, \vec{r}_2) = \frac{8}{\pi a_0^3} e^{-2(r_1+r_2)/a_0}$$

$$E_0 = -13.6 \frac{z^2}{n^2} \text{ eV} \quad , \quad z=2, n=1, E_1 = -13.6 \text{ eV}$$

$$\begin{aligned} \text{Total ground state energy } E_{\text{He}} &= 2(-13.6) z^2 \text{ eV} \\ &= -13.6 \times 8 \text{ eV} \\ &= -109 \text{ eV} \end{aligned}$$

We apply variation principle with trial function

$$\psi_0 = \frac{8}{\pi a_0^3} e^{-2(r_1 + r_2)/a_0}$$

We need to calculate the expectation value $\langle \psi_0 | \hat{H} | \psi_0 \rangle$

$$\hat{H} = \hat{H}_0 + \hat{V}_{ee}$$

$$\text{where } \hat{H}_0 = -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{2e^2}{r_1} - \frac{2e^2}{r_2}$$

$$\text{and } \hat{V}_{ee} = \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}$$

$$\hat{H}_0 \psi_0 = 8E_1 \psi_0$$

$$\hat{H} \psi_0 = (\hat{H}_0 + \hat{V}) \psi_0 = (8E_1 + V_{ee}) \psi_0$$

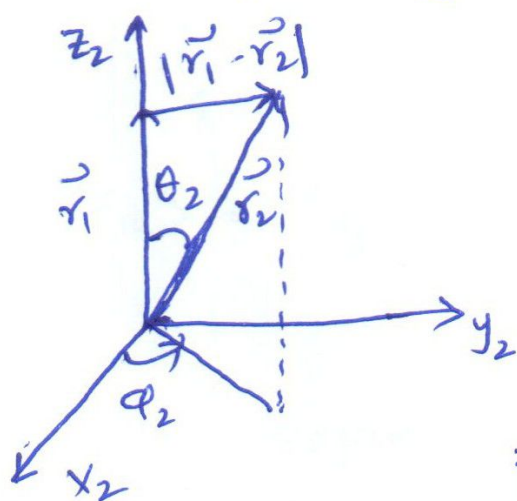
$$\langle \psi_0 | \hat{H} | \psi_0 \rangle = 8E_1 \langle \psi_0 | \psi_0 \rangle + \langle \psi_0 | V_{ee} | \psi_0 \rangle$$

$$\langle \hat{H} \rangle = 8E_1 + \langle V_{ee} \rangle$$

$$\langle V_{ee} \rangle = \langle \psi_0 | V_{ee} | \psi_0 \rangle$$

$$= e^2 \left(\frac{8}{\pi a_0^3} \right)^2 \int \frac{e^{-4(r_1 + r_2)/a_0}}{|\vec{r}_1 - \vec{r}_2|} d^3\vec{r}_1 d^3\vec{r}_2$$

To evaluate the integral, we carry out co-ordinate transformation $\rightarrow$ orient the co-ordinate system so that $\vec{r}_1$ is along z-axis.



$$|\vec{r}_1 - \vec{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}$$

We do the integral over $\vec{r}_2$ first.

$$I_2 = \int \frac{e^{-4r_2/a_0}}{|\vec{r}_1 - \vec{r}_2|} d^3r_2$$

$$= \int \frac{e^{-4r_2/a_0} r_2^2 \sin \theta_2 dr_2 d\theta_2 d\phi_2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}}$$

Integral over ϕ_2 gives 2π

Integral over θ_2

$$\begin{aligned} & \int_0^\pi \frac{\sin \theta_2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}} d\theta_2 \\ &= \left[\frac{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}}{r_1 r_2} \right]_0^\pi \\ &= \frac{1}{r_1 r_2} \left[\sqrt{r_1^2 + r_2^2 + 2r_1 r_2} - \sqrt{r_1^2 + r_2^2 - 2r_1 r_2} \right] \\ &= \frac{1}{r_1 r_2} \left[(r_1 + r_2) - |r_1 - r_2| \right] \\ &= \frac{1}{r_1 r_2} \left[\begin{array}{l} r_1 + r_2 - r_1 + r_2 \\ r_1 + r_2 + r_1 - r_2 \end{array} \right] \left\{ \begin{array}{l} r_2 < r_1 \\ r_2 > r_1 \end{array} \right. \\ &= \frac{2}{r_1} \left\{ \begin{array}{l} r_2 < r_1 \\ r_2 > r_1 \end{array} \right. \end{aligned}$$

$$I_2 = 4\pi \left[\int_0^{r_1} e^{-4r_2/a_0} r_2^2 \left(\frac{1}{r_1}\right) dr_2 + \int_{r_1}^\infty e^{-4r_2/a_0} r_2^2 \left(\frac{1}{r_2}\right) dr_2 \right]$$

$$= \frac{\pi a_0^3}{8r_1} \left[1 - \left(1 + \frac{2r_1}{a_0}\right) e^{-4r_1/a_0} \right]$$

$$\Rightarrow \langle V_{ee} \rangle = e^2 \left(\frac{8}{\pi a_0^3} \right) \int \left[1 - \left(1 + \frac{2r_1}{a_0}\right) e^{-4r_1/a_0} \right] r_1 \sin \theta_1 dr_1 d\theta_1 d\phi_1$$

Angular integrals give 4π

Integral over r_1 gives $\frac{5a_0^3}{128}$

$$\langle V \rangle = \frac{5}{4a_0} \left(\frac{e^V}{\cancel{4\pi}} \right) = + \frac{5}{2} \cdot \overbrace{\left(+13.6 \right)}^{|E_1| \frac{e^V}{2a_0} = 13.6 \text{ eV}} = +34 \text{ eV}$$

$$\langle H \rangle = -109 + 34 = -75 \text{ eV}$$

close to experimental value. We can do better. Instead of putting $z=2$, we take z as a parameter in the trial wave function.

$$\psi_0 = \frac{z^3}{\pi a_0^3} e^{-z(r_1+r_2)/a_0}$$

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2)}_{\hat{H}_0} - e^V \left(\frac{z}{r_1} + \frac{z}{r_2} \right) + \underbrace{e^V \left(\frac{z-2}{r_1} + \frac{z-2}{r_2} \right)}_{\hat{H}_1} + \underbrace{\frac{e^V}{|\vec{r}_1 - \vec{r}_2|}}_{V_{ee}}$$

$\hat{H}_0 \rightarrow$ corresponding eigen function ψ_0

Note: the Hamiltonian is unchanged.

We added and subtracted $e^V \left(\frac{z}{r_1} + \frac{z}{r_2} \right)$

$$\langle \psi_0 | H | \psi_0 \rangle = \langle \psi_0 | \hat{H}_0 | \psi_0 \rangle + \langle \psi_0 | \hat{H}_1 | \psi_0 \rangle + \langle \psi_0 | \hat{V}_{ee} | \psi_0 \rangle$$

$$\langle \psi_0 | \hat{H}_0 | \psi_0 \rangle = \langle \psi_0 | E_0 | \psi_0 \rangle = 2E_1 z^V \langle \psi_0 | \psi_0 \rangle$$

$$\langle \psi_0 | \hat{H}_1 | \psi_0 \rangle = \langle \psi_0 | \frac{e^V(z-2)}{r_1} | \psi_0 \rangle + \langle \psi_0 | \frac{e^V(z-2)}{r_2} | \psi_0 \rangle$$

$$\psi_0 | \hat{H}_1 | \psi_0 \rangle$$

$$= e^{\nu}(z-2) \int \int \psi_{100}^*(\vec{r}_1) \psi_{100}^*(\vec{r}_2) \frac{1}{r_1} \psi_{100}(\vec{r}_1) \psi_{100}(\vec{r}_2) d^3\vec{r}_1 d^3\vec{r}_2$$

$$+ e^{\nu}(z-2) \int \int \psi_{100}^*(\vec{r}_1) \psi_{100}^*(\vec{r}_2) \frac{1}{r_2} \psi_{100}(\vec{r}_1) \psi_{100}(\vec{r}_2) d^3\vec{r}_1 d^3\vec{r}_2$$

$$= e^{\nu}(z-2) \left[\int \psi_{100}^*(\vec{r}_1) \frac{1}{r_1} \psi_{100}(\vec{r}_1) d^3\vec{r}_1 \int \psi_{100}^*(\vec{r}_2) \psi_{100}(\vec{r}_2) d^3\vec{r}_2 \right. \\ \left. + \int \psi_{100}^*(\vec{r}_1) \psi_{100}(\vec{r}_1) d^3\vec{r}_1 \int \psi_{100}^*(\vec{r}_2) \frac{1}{r_2} \psi_{100}(\vec{r}_2) d^3\vec{r}_2 \right]$$

$$= e^{\nu}(z-2) \left[\langle \psi_{1s} | \frac{1}{r} | \psi_{1s} \rangle + \langle \psi_{1s} | \frac{1}{r} | \psi_{1s} \rangle \right]$$

$$= 2e^{\nu}(z-2) \langle \frac{1}{r} \rangle_{1s}$$

$$\psi_{1s} = \sqrt{\frac{z^3}{\pi a_0^3}} e^{-zr/a_0}, \quad a_0 = \frac{\hbar^2}{m e v}$$

$$\langle \frac{1}{r} \rangle_{1s} = \frac{z^3}{\pi a_0^3} \int_0^{\infty} e^{-2zr/a_0} \frac{1}{r} r^2 dr \int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\phi$$

$$= \frac{4\pi z^3}{\pi a_0^3} \int_0^{\infty} e^{-2zr/a_0} r dr$$

$$\underbrace{\hspace{10em}}_{\frac{a_0^2}{4z^2}}$$

$$= \frac{4z^3}{a_0^3} \cdot \frac{a_0^2}{4z^2} = \frac{z}{a_0} \Rightarrow \langle \psi_0 | \hat{H}_1 | \psi_0 \rangle = \frac{2e^{\nu}(z-2)}{z/a_0}$$

Again $\langle V_{ee} \rangle = \frac{5e^{\nu}}{4a_0}$ for $z=2$, $\psi_0 = \frac{8}{\pi a_0^3} e^{-2(r_1+r_2)/a_0}$

For unknown z , new trial function $\psi_0 = \frac{z^3}{\pi a_0^3} e^{-z(r_1+r_2)/a_0}$

$$\Rightarrow \frac{2}{a_0} \rightarrow \frac{2}{2a_0/z} \Rightarrow a_0 \rightarrow \frac{2a_0}{z}$$

$$\langle V_{ee} \rangle = \frac{5}{4a_0} \cdot \frac{z}{2}, \quad e^{\nu} = \frac{5z}{8a_0} e^{\nu}$$

$$\langle H \rangle = \langle H_0 \rangle + \langle \psi_0 | H_1 | \psi_0 \rangle + \langle V_{ee} \rangle$$

$$= 2E_1 z^2 + 2(z-2)z \frac{e^2}{a_0} + \frac{5z}{8} \frac{e^2}{a_0}$$

$$= 2E_1 z^2 + 4(z-2)z(-E_1) + \frac{5z}{4}(-E_1) \quad (\because E_1 = -\frac{e^2}{2a_0})$$

$$= E_1 \left(2z^2 - 4z^2 + 8z - \frac{5}{4}z \right)$$

$$= E_1 \left(-2z^2 + \frac{27}{4}z \right)$$

Minimize $\langle H \rangle$ w.r.t. $z \Rightarrow$

$$\frac{d}{dz} \langle H \rangle = -4z + \frac{27}{4} = 0$$

$$\Rightarrow z = \frac{27}{16} = 1.69 \quad *$$

$$\langle H \rangle = E_1 \left(-2z^2 + \frac{27}{4}z \right)$$

$$= -13.6 \left[-2 \cdot (1.69)^2 + \frac{27}{4} \cdot 1.69 \right]$$

$$= -77.5 \text{ eV}$$

Expt value -79.0 eV .

- * $z = 1.69 < 2$ suggests that the nucleus charge is partially screened by other electron, thus reducing the effective charge from ~~to~~ 2 down to 1.69.